

WELLSITE INTERPRETATION & REAL-TIME MONITORING

Mud Logging • Formation Evaluation • Drilling Surveillance

LIVE DRILLING DATA

GEOLOGY + GAS

ALARMS & DECISIONS

PT. GEO OPTI OILFIELD

Technical Training / Operations Presentation

Purpose & Operational Value

Integrate geology, gas, drilling and mud-system data into one operational picture.

Identify formation changes, hydrocarbon shows and pressure-related trends while drilling.

Support early recognition of kick/loss indicators, hole-cleaning issues and drilling dysfunction.

Provide clear, time-aligned communication to Company Man, Drilling, Geology and remote teams.

Wellsite Interpretation Workflow



Real-Time Data Architecture



Core Parameters to Monitor

Drilling

ROP, WOB, RPM, Torque, SPP, SPM

Hydraulics

Flow in/out, pits, trip tank, mud weight, ECD trend

Gas

Total Gas, C1–C5, BG, connection/trip gas

Geology

Lithology %, cavings, fluorescence, formation tops

Derived

d-Exponent, MSE, lag, hole-cleaning indicators

Geological Interpretation at Wellsite

Correlate cuttings lithology with ROP, gas response and available LWD/MWD curves.

Pick formation tops and boundaries from consistent changes rather than one parameter alone.

Track cavings, grain size, sorting, cement, accessory minerals and alteration.

Distinguish drilled formation response from operational artifacts and lag effects.

Update interpreted lithology continuously as new samples and curves become available.

Cuttings & Formation Evaluation

CUTTINGS DESCRIPTION

- Rock type & percentage
- Color / texture / grain size
- Hardness & cement
- Porosity indication
- Cavings / contamination
- Calcimetry / shale density as applicable

HYDROCARBON SHOW

- Fluorescence
- Cut / solvent response
- Odor / stain
- Gas association
- Show intensity & interval
- Correlation with ROP / lithology

INTERPRETATION

- Formation top
- Reservoir quality indication
- Seal / shale interval
- Fracture / permeability clues
- Pressure-related cavings
- Sample quality confidence

Gas Data Interpretation

Establish stable Background Gas (BG) before interpreting peaks.

Differentiate drilled/liberated gas, produced gas, recycled gas and contamination gas.

Evaluate Total Gas together with C1–C5 composition and gas ratios.

Connection and trip gas require correct lag/bottom-up timing and operational context.

A gas increase is interpreted with ROP, flow, pits, mud weight, lithology and pressure indicators.

Lag Time & Depth Control

WHY LAG MATTERS

Gas and cuttings observed at surface must be assigned to the correct drilled depth. Incorrect lag shifts formation tops, gas shows and pressure interpretation.

VALIDATION

Pump strokes / flow
Annular volume
Tracer or operational marker
Cuttings arrival
Periodic recalculation

CONTROL

Update after hole-size, BHA, flow-rate or mud-property changes. Record both time-based and depth-based events.

Pore Pressure Surveillance

Track d-Exponent / corrected d-Exponent trend by formation and bit run.

Correlate with ROP, WOB, torque, shale density/cavings, gas and mud-weight behavior.

Look for trend reversal or departure from established normal compaction trend.

Use multiple independent indicators before escalating abnormal-pressure concern.

Document interpretation confidence and operational recommendation.

Kick / Influx Early Warning

PRIMARY INDICATORS

- Pit gain
- Flow-out > expected
- Flow with pumps off
- Increasing connection gas
- Unexpected drilling break
- Mud-weight / return changes

CONFIRMATION

- Verify sensor / pit transfer
- Check pumps & surface operations
- Compare trip tank / flow balance
- Confirm lag-related gas
- Inform Driller / Company Man

RESPONSE SUPPORT

Mud logging provides early surveillance and communication. Well-control actions remain under the rig's approved well-control procedures and responsible drilling personnel.

Loss Circulation Monitoring

Monitor active pits, trip tank, flow-in/flow-out and return behavior continuously.

Classify seepage, partial or total losses using operationally agreed thresholds.

Correlate losses with lithology, fractures, drilling break, torque and pressure changes.

Record LCM sweeps, mud transfers and dilution to avoid false alarms.

Trend cumulative loss versus depth/time for decision support.

Hole Cleaning & Stuck Pipe Risk

Trend torque, drag/hookload behavior, SPP, flow, ROP and cuttings volume.

Watch for increasing cavings, poor returns, pack-off signatures and abnormal overpull.

Use connection trends and bottoms-up observations to assess cleaning efficiency.

Separate mechanical restriction, differential sticking and solids-bed symptoms.

Escalate sustained multi-parameter deterioration before a stuck-pipe event develops.

Drilling Performance – ROP & MSE

ROP INTERPRETATION

ROP responds to formation strength, WOB, RPM, hydraulics, bit condition and pressure. Use normalized context before calling a formation change.

MSE INTERPRETATION

Mechanical Specific Energy helps compare drilling efficiency. Rising MSE with falling ROP can indicate inefficiency, bit wear, dysfunction or harder formation.

Connection / Trip / Swab Gas

Connection gas: evaluate at the expected bottoms-up/lag position after connection.

Trip gas: evaluate after returning to bottom and circulating the trip interval.

Swab gas: may reflect transient pressure reduction caused by pipe movement.

Recycled gas: generally repeats a prior gas response after a circulation cycle and may have altered composition.

Always compare gas shape, magnitude, timing and operational history.

Real-Time Composite Log

Depth-aligned integration: geology + gas + drilling + interpretation

DEPTH	LITHOLOGY	TOTAL GAS	ROP	PRESSURE	REMARKS
2450	SHALE	BG 8u	18	d-Exp ↓	Top reservoir
2475	SST	▲ 42u	42	CG ↑	Gas show
2500	SST/SH	▲ 65u	55	MW check	Verify pressure
2525	SHALE	BG 12u	22 m/h	Trend stable	Communicate

Remote Monitoring Dashboard

LIVE VIEW

Time & depth plots
Current operation state
Gas + drilling + mud
Well trajectory / status

ANALYTICS

Historical playback
Trend comparison
Derived channels
Event tagging / alarms

REMOTE COLLABORATION

Rig-to-office visibility
Geology & drilling review
Faster escalation
Shared operational picture

Alarm & Escalation Philosophy

ADVISORY

Single parameter deviation — verify data and context.

WARNING

Confirmed trend or multi-parameter anomaly — inform relevant discipline.

CRITICAL

High-confidence safety/operational risk — immediate escalation per procedure.

Wellsite Communication Workflow

Mud Logger validates sensor/data quality and identifies the event.

Senior Mud Logger / Wellsite Geologist interprets geological and gas significance.

Driller and Company Man receive operationally relevant alarms immediately.

Mud Engineer / Directional / MWD-LWD teams are engaged when their data are required.

Remote operations team receives synchronized plots, event notes and recommended follow-up.

Daily Deliverables & Reporting

Real-Time

Live dashboard, alarms, event log

Daily

Daily mud logging report, gas summary, lithology, drilling events

Depth Log

Composite mud log / master log, formation tops, shows

End of Well

Final log, interpretation summary, lessons learned, data archive

Recommended KPI Dashboard

KPI 1

Data availability / sensor uptime

KPI 2

Depth & lag validation status

KPI 3

Alarm response time

KPI 4

Gas-system calibration compliance

KPI 5

Formation-top / sample coverage

KPI 6

NPT / drilling dysfunction events

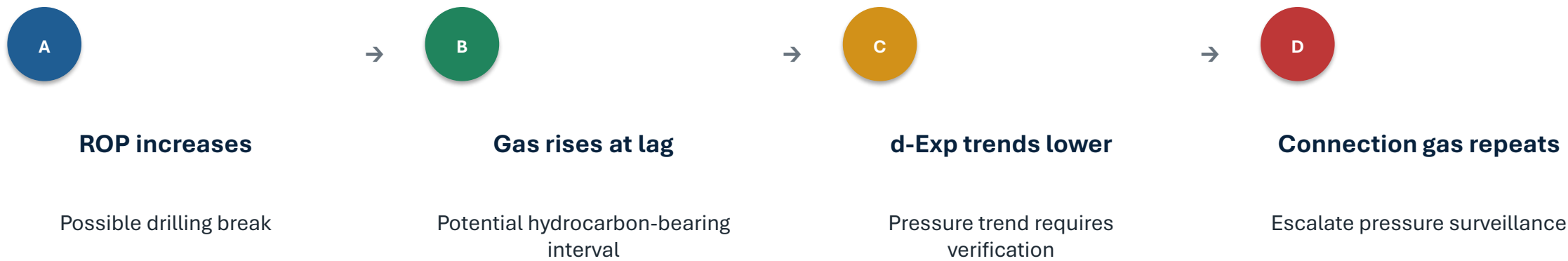
KPI 7

ROP / MSE by section or bit

KPI 8

Loss / gain event tracking

Example Interpretation Scenario



Interpretation is strengthened when independent indicators converge and operational artifacts have been excluded.

Data Quality / QC Checklist

Sensor zero/span and calibration status verified.

Gas trap position, gas-line flow and chromatograph response checked.

Depth synchronization and lag time validated.

Pit transfers, sweeps, LCM, dilution and rig operations correctly annotated.

Time/depth plots reviewed for spikes, flat-lines, drift and communication gaps.

Interpretations carry confidence level and supporting parameters.

Conclusion

Real-time monitoring is most valuable when data acquisition, QC and interpretation are integrated.

Wellsite interpretation should combine geology, gas, drilling, hydraulics and operational context.

Trend-based, multi-parameter surveillance reduces false alarms and improves early recognition.

Remote visualization and WITS/WITSML workflows enable faster collaboration and consistent reporting.

PT. Geo Opti Oilfield can structure the workflow from rig sensors to actionable wellsite intelligence.